REPORT OF

HOUSE PRICE PREDICTION IN RWANDA WITH MULTIPLE LINEAR REGRESSION

Module: Machine Learning

Department: Electrical and Electronics Department

Option and Class: ETT Y4 BTech

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1. Introduction and Objective

This project develops a machine learning model to predict house prices in Rwanda. The model uses house characteristics such as area, bedrooms, bathrooms, age, distance from the city centre, parking spaces, and neighbourhood.

2. Data Exploration and Cleaning

The dataset was loaded and inspected using Python, Pandas, and NumPy. Missing values, duplicate records, and unusual values were checked and handled. The categorical variable **Neighborhood** was converted using one-hot encoding..

Cleaning decisions:

Problem found	Where	Action taken	Reason
Missing numeric values	Numeric columns	Median imputation	Preserves observations and is less sensitive to extreme values.
Missing Neighborhood	Neighborhood	Mode imputation	Uses the most frequent observed category.
Exact duplicate records	House_ID 1011, 1026 and 1041	Remove exact duplicates	Prevents repeated observations receiving extra weight.
IQR outliers	Numeric variables and target	Cap at IQR fences	The dataset is small and there is no domain evidence that the extreme values are errors.
Identifier	House_ID	Exclude from model	An identifier is not a meaningful price predictor.

3. Exploratory Visualisation

Visualizations were created to understand price distribution, relationships between area and price, neighbourhood prices, and correlations between variables.

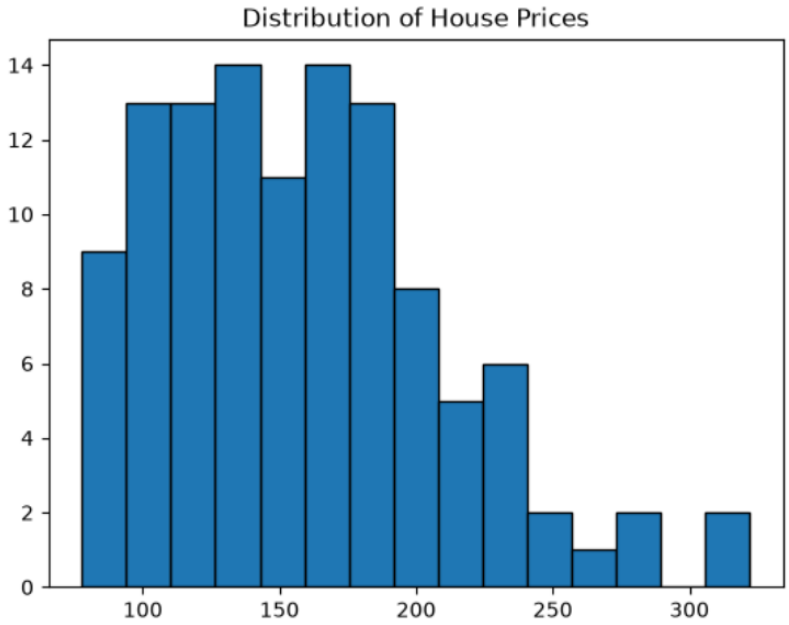


Figure 1. Distribution of house prices. The chart shows the spread of the cleaned target values.



Figure 2. Area versus house price. The plot is used to assess the relationship between house size and price.

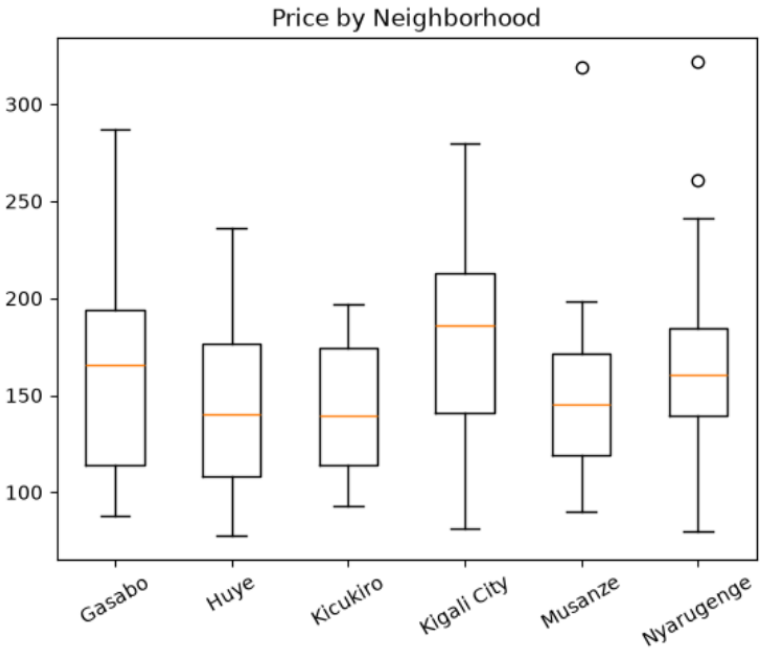


Figure 3. House price by neighborhood. The distributions show differences in prices across locations.

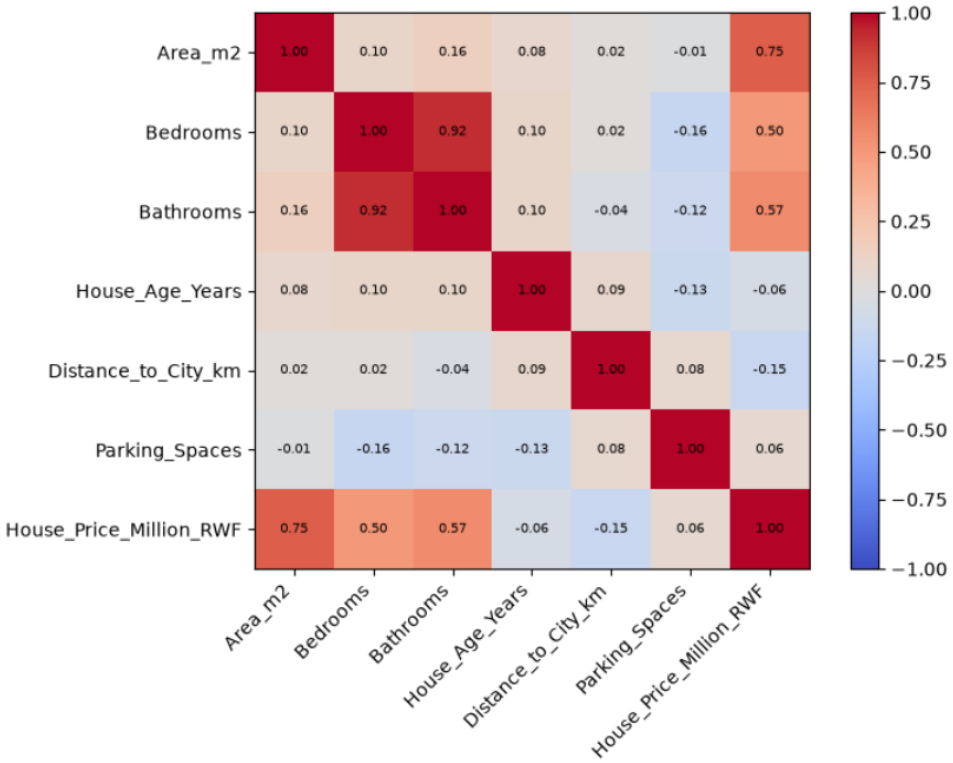


Figure 4. Correlation heatmap. It helps identify relationships among numeric variables and possible multicollinearity.

4. Building the Multiple Linear Regression Model

A **Multiple Linear Regression** model was developed using Scikit-learn. The data was divided into **80% training and 20% testing** using `random_state=42`.

5. Evaluation and Interpretation

Dataset	R2	Adjusted_R2	MAE_Million_RWF	RMSE_Million_RWF
Training	Training	0.881241	0.864493	14.493915
Testing	0.787919	0.575839	18.375411	22.936607

The Actual versus predicted prices and residual plots were also used to assess model performance and regression assumptions.

6. Diagnostic Analysis and Assumptions

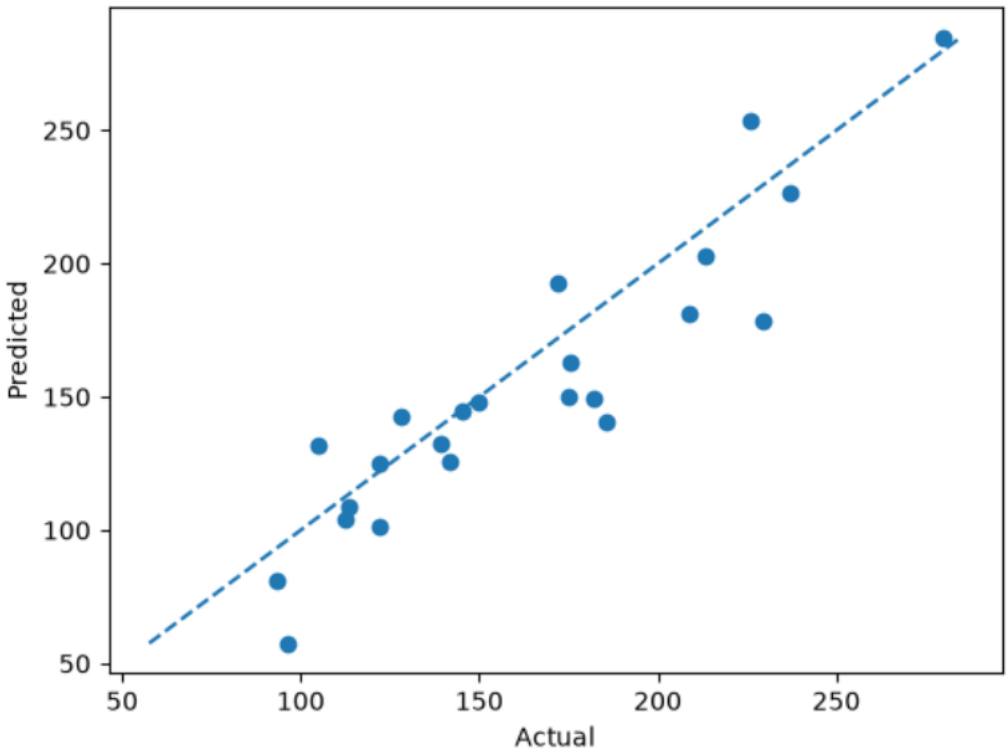


Figure 5. Actual versus predicted prices for the test set.

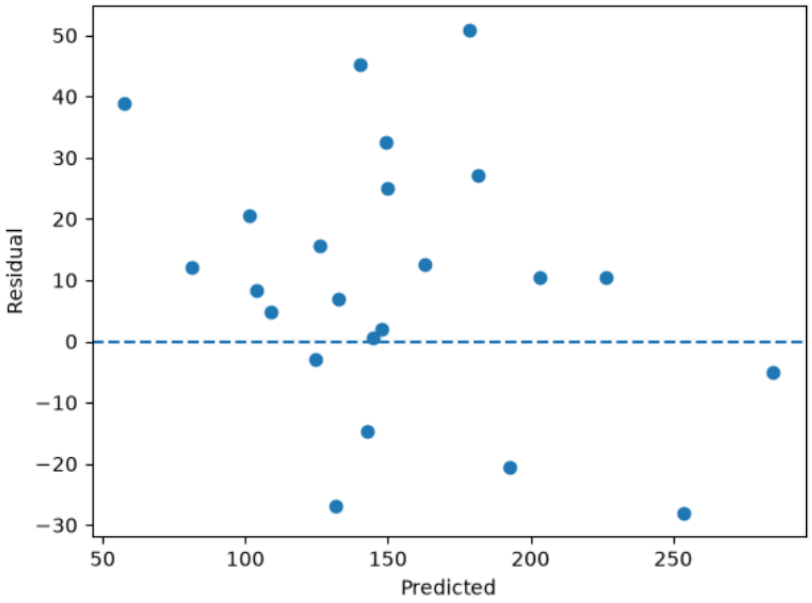


Figure 6. Residuals versus predicted values.

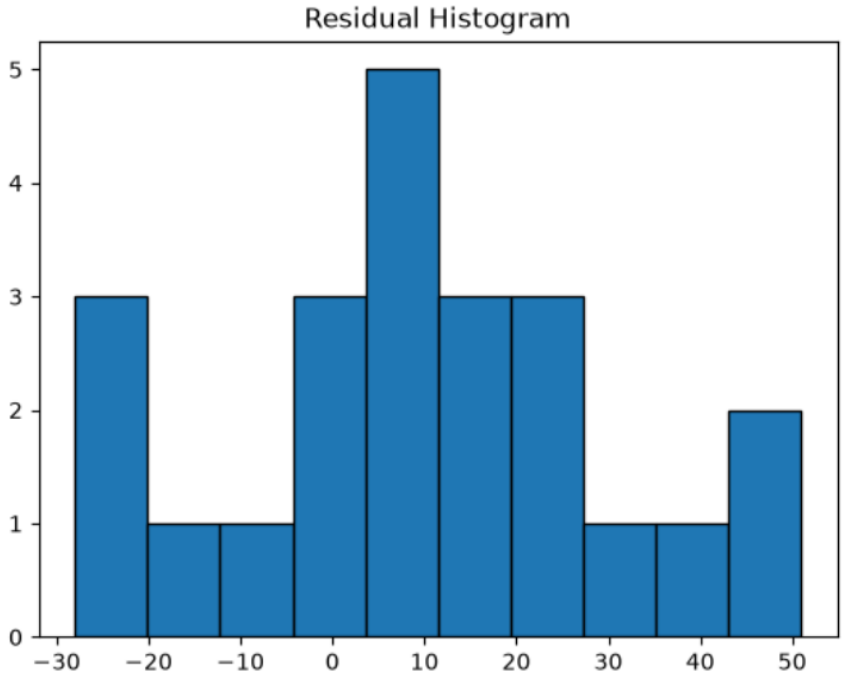


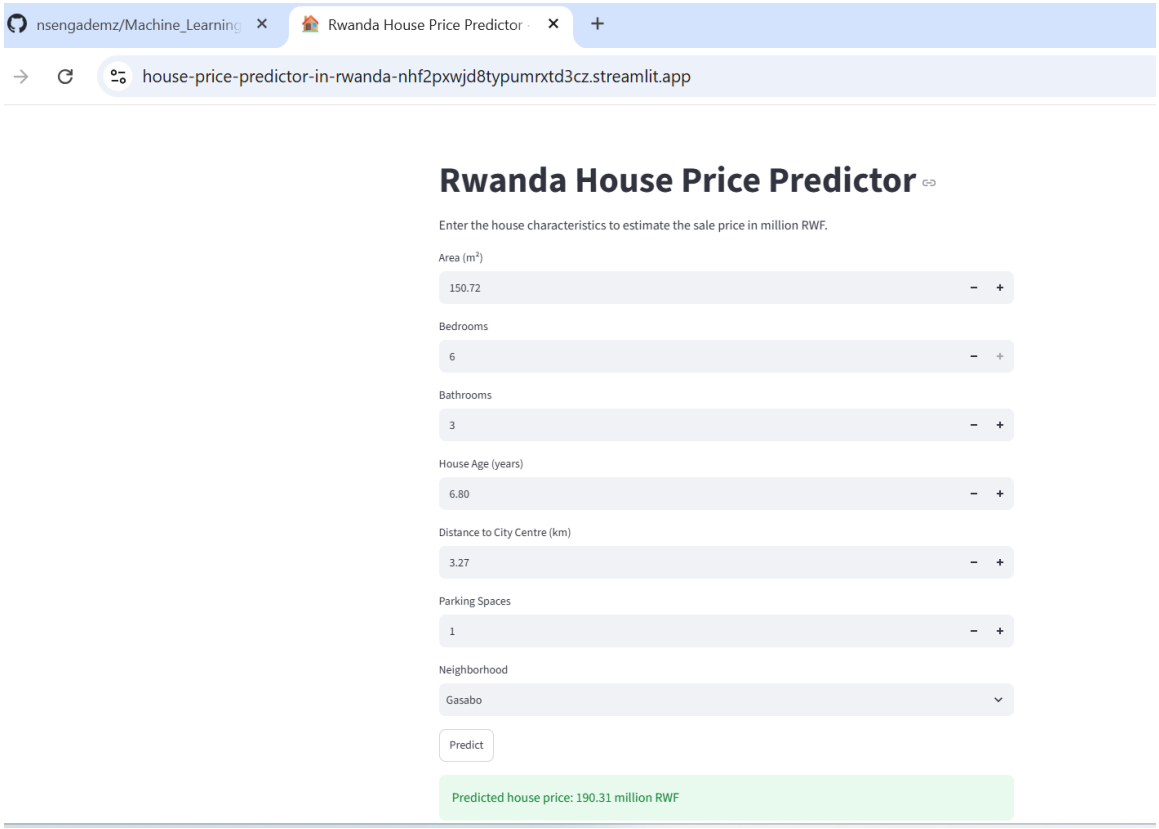
Figure 7. Distribution of test residuals.

7. Streamlit Application

The Streamlit application is provided in app.py. It loads house_price_model.sav with @st.cache_resource, uses a relative file path, accepts all seven predictor inputs, validates values against the cleaned training ranges, and displays the predicted price in million RWF.

The application was tested locally and prepared for online deployment.

Now, Sreamlit is already to be viewed in browser



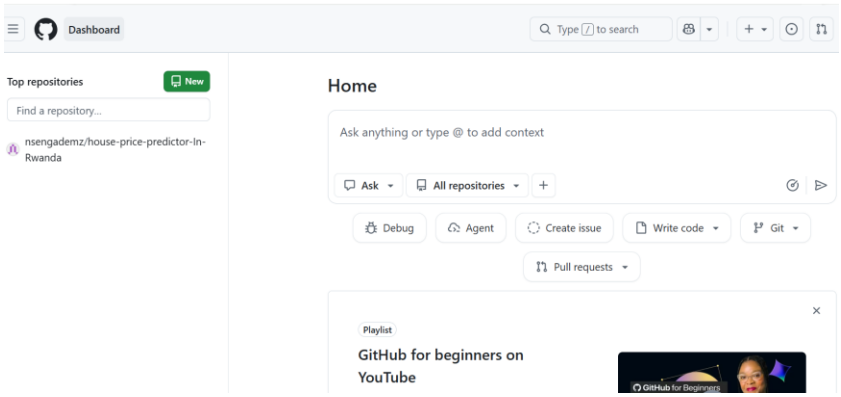
8. GitHub and Online Deployment

Public GitHub repository created,

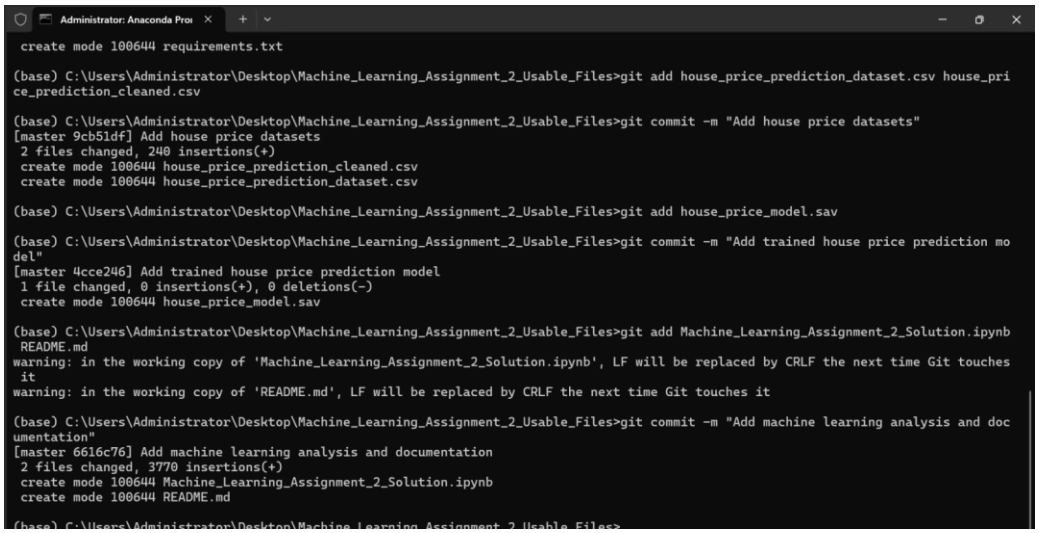
The project source code and required files were uploaded to GitHub:

Repository:
https://github.com/nsengademz/Machine_Learning_Assignment_2_Usable_Files

The repository contains the project files, trained model, notebook, and application code.



At least these are four meaningful commits with clear messages,



9. Conclusion

The project demonstrates how machine learning can be used to estimate house prices from property characteristics. The model and Streamlit application provide a simple tool for generating house price predictions.

10. Limitations

The dataset is relatively small, and linear regression may not capture all factors affecting house prices. Additional data and more advanced models could improve prediction accuracy.